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Procedural Fairness Failures in RLHF from Preference Averaging

Standard RLHF Violates Procedural Fairness by Drowning Minority Preference Signals in Aggregated Reward Models; PA-RLHF Separates Optimization Across Preference Modes

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Reinforcement Learning from Human Feedback (RLHF) aggregates heterogeneous annotator preferences into a single reward model, treating all annotators as draws from one distribution. A new paper shows this induces a *procedural fairness failure*: majority preference groups dominate reward learning while minority preferences are systematically under-represented. Preference-Aware RLHF (PA-RLHF) remedies this by separating reward optimization across preference modes at the reward learning stage.

AT A GLANCE

Problem: Standard RLHF averages heterogeneous preferences into a single reward model, causing minority preference groups to be under-represented.

Why it matters: As LLMs are deployed across diverse populations, preference homogeneity is a poor assumption — and its failure is invisible to standard metrics.

Method: Define procedural fairness for alignment; prove standard RLHF violates it; propose PA-RLHF with per-group reward models.

Result: PA-RLHF improves procedural fairness metrics while maintaining outcome quality (win rates, helpfulness) across all tested scenarios.

Evidence: Gains largest when inter-group preference divergence and population size imbalance are greatest.

The Big Picture

Human preferences are not homogeneous. Annotators employed in RLHF pipelines come from different cultural backgrounds, hold different values, and have different conceptions of what constitutes helpful, harmless, and honest AI behavior. Standard RLHF treats all annotator preferences as if they were noisy draws from a single underlying distribution, fitting a single aggregated reward model r_θ to the pooled data.

When preferences are genuinely multi-modal — when subgroups of annotators hold coherently different values — this aggregation is not merely imprecise. It actively *dilutes* minority preference signals in proportion to the size of the minority. Smaller groups contribute less to the aggregate loss, so the reward model learns to satisfy the majority and discard the minority. The paper calls this a **procedural fairness failure**: the process is unfair to minority groups, independent of whether the resulting policy happens to satisfy them by coincidence.

Why Existing Approaches Fall Short

Prior work on fairness in RLHF has focused on **outcome fairness**: ensuring the final policy produces acceptable outputs for all subgroups. But outcome fairness can be achieved by coincidence (a policy that satisfies the majority may also happen to satisfy the minority on some tasks) and can be gamed by optimizing for population-level statistics. Outcome fairness does not address whether the reward learning *process* treats all groups equitably.

Existing personalized preference methods (per-user reward models, preference embeddings) require per-user data at inference time and scale poorly. PA-RLHF operates at the group level during reward training, requiring no per-user data after deployment.

Core Mechanism

Let $\mathcal{A}$ be the annotator pool, partitioned into K preference groups $G_1, \ldots, G_K$ with group-specific preference distributions $\mathcal{D}_1, \ldots, \mathcal{D}_K$. The key steps of PA-RLHF are:

1. **Preference clustering:** Identify latent groups G_k using embedding-based clustering of annotator preference vectors or label co-occurrence patterns.
2. **Per-group reward modeling:** Fit a separate reward model r_{θ_k} for each group G_k using only that group's preference data.
3. **Fairness-weighted policy optimization:** Combine per-group reward signals during PPO training using a fairness weighting w_k that does not simply equal the group's population share $\alpha_k = |G_k|/|\mathcal{A}|$.

The fairness weighting w_k is the key design choice: uniform weighting ($w_k = 1/K$) maximizes procedural equity; intermediate weightings balance procedural and outcome fairness objectives.

Technical Formulation

Standard RLHF fits:

$$r_\theta = \arg\min_\theta \mathbb{E}_{(x,y_w,y_l)\sim\mathcal{D}}[-\log \sigma(r_\theta(x,y_w) - r_\theta(x,y_l))]$$

where $\mathcal{D} = \sum_k \alpha_k \mathcal{D}_k$ is the mixture distribution.

PA-RLHF fits K per-group reward models:

$$r_{\theta_k} = \arg\min_{\theta_k} \mathbb{E}_{(x,y_w,y_l)\sim\mathcal{D}_k}[-\log \sigma(r_{\theta_k}(x, y_w) - r_{\theta_k}(x, y_l))]$$

and optimizes the policy with the fairness-weighted reward:

$$r_{\text{fair}}(x, y) = \sum_{k=1}^{K} w_k \cdot r_{\theta_k}(x, y), \quad \sum_k w_k = 1$$

Procedural fairness is defined as: the gradient contribution of group G_k to the aggregate reward model should be proportional to its "representation right" rather than its population size. Formally, standard RLHF satisfies this only when $\alpha_k = 1/K$ (equal group sizes), which is rarely the case in practice.

Learning or Inference Procedure

PA-RLHF requires one additional step versus standard RLHF:

1. **Cluster annotators** into K groups (one-time offline step).
2. **Train K reward models** in parallel (same cost as standard training when using parameter-efficient methods; $K\times$ cost otherwise).
3. **Run PPO** with r_{fair} as the reward signal.

Inference is identical to standard RLHF — no per-user data is required.

What the Guarantee Says

The paper proves that standard RLHF is procedurally unfair whenever annotator group sizes differ ($\alpha_k \neq 1/K$) and group preferences are non-identical. PA-RLHF with uniform weighting ($w_k = 1/K$) is the unique solution that achieves procedural fairness under this definition. No guarantee is provided for outcome fairness — PA-RLHF may improve or maintain outcome metrics, but cannot guarantee it.

Experimental Findings

- **Procedural fairness:** PA-RLHF improves procedural fairness metrics across all tested scenarios with heterogeneous annotator groups.
- **Outcome quality:** Win rates and helpfulness metrics are maintained or slightly improved relative to the standard RLHF baseline.
- **Group imbalance sensitivity:** Gains grow with group size imbalance ratio — the regime most common in real crowdsourced annotation datasets.
- **Preference divergence sensitivity:** Gains also grow with inter-group preference divergence, confirming the mechanism targets the right failure mode.

Ablations and Interpretation

How many groups K? The paper tests $K \in \{2, 3, 5\}$; fairness gains are present for all values, with $K = 3$ providing the best balance of fairness and computation overhead.

Fairness weighting w_k: Uniform ($w_k = 1/K$) maximizes procedural fairness but can hurt outcome quality in extreme scenarios. Population-proportional ($w_k = \alpha_k$) recovers standard RLHF. An intermediate value provides the best empirical outcome-fairness tradeoff.

What about outcome fairness? The paper notes that procedural fairness and outcome fairness are not equivalent: PA-RLHF improves procedural fairness without guaranteeing outcome fairness, and a system can achieve outcome fairness while violating procedural fairness. Both matter and both need to be measured.

Reference

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